

Lab 5

better late than never

Your Name

We should have done this weeks ago, but JZ is an idiot. Please forgive him.

Basic math

```
# TBD
```

Vectors

Subsetting

Create a vector containing the numbers 6, π , $\sqrt{2}$, e , and 0:

```
# TBD
```

Center-of-mass example

```
# TBD
```

Binary string example

```
# TBD
```

Sampling

Flip a fair coin ten times

```
# TBD
```

Roll a fair die ten times

```
# TBD
```

Deal a five card hand from a shuffled deck

```
deck_of_cards <- c("AH", "2H", "3H", "4H", "5H", "6H", "7H", "8H", "9H", "10H", "JH", "QH", "KH",  
                  "AD", "2D", "3D", "4D", "5D", "6D", "7D", "8D", "9D", "10D", "JD", "QD", "KD",  
                  "AC", "2C", "3C", "4C", "5C", "6C", "7C", "8C", "9C", "10C", "JC", "QC", "KC",  
                  "AS", "2S", "3S", "4S", "5S", "6S", "7S", "8S", "9S", "10S", "JS", "QS", "KS",  
                  "A", "2", "3", "4", "5", "6", "7", "8", "9", "10", "J", "Q", "K")  
  
# TBD
```

Center-of-mass example again

Simulate the center-of-mass distribution and verify that the empirical proportions are close to the actual probabilities:

```
range_x <- -6:2  
masses <- c(3, 7, 6, 4, 4, 3, 3, 2, 1)  
pmf <- masses / sum(masses)  
  
# TBD
```

Plotting

Plot a pmf for the center-of-mass example

```
# TBD
```

Plot from PSET 0

Plot this function from 2 to 15:

$$f(x) = \frac{1}{x(\ln x)^{-1}}.$$

```
# TBD
```

Plot from Lab 4

For fixed $f_- = f_+ = 0.2$, plot these functions:

$$h(p) = \frac{(1 - f_-)p}{(1 - f_-)p + f_+(1 - p)}, \quad g(p) = \frac{p}{p + f_+}, \quad 0 \leq p \leq 1.$$

```
# TBD
```

Control flow and logic

TBD

Binomial practice

Plot the pmf

Plot the pmf of $X \sim \text{Binom}(n = 6, p = 0.4)$:

```
# TBD
```

Compute a probability

If $X \sim \text{Binom}(n = 25, p = 0.0162)$, compute $P(X > 0)$:

```
# TBD
```

Simulate and verify

Simulate $X \sim \text{Binom}(n = 10, p = 0.8)$ several times and check if the empirical proportions match the theoretical probabilities.

```
# TBD
```

Poisson practice

Plot the pmf

Plot the pmf of $X \sim \text{Poisson}(\lambda = 100)$:

```
# TBD
```

Compute a probability

Let $X \sim \text{Poisson}(\lambda = 100)$ be the number of claims that an insurance company receives in a month. What is the probability that they receive an above-average number of claims?

```
# TBD
```

If $X \sim \text{Poisson}(\lambda = 4)$, what is the probability that X is even?

```
# TBD
```

Simulate and verify

Simulate $X \sim \text{Poisson}(\lambda = 100)$ a bunch and verify that the empirical proportions are close to the true probabilities.

```
# TBD
```

LaTeX practice

Binomial pmf

Typeset the formula for the binomial pmf:

TBD

Poisson pmf

Typeset the formula for the Poisson pmf:

TBD

Gamma function

Typeset the formula for the gamma function (Problem Set 0):

TBD

Funky expected value

Typeset the

TBD